

Thermochemical waste-heat recuperation by steam methane reforming with flue gas addition

Dmitry Pashchenko 

Samara State Technical University,
Samara, Russia

Correspondence

Dmitry Pashchenko, Samara State
Technical University, Samara, Russia.
Email: pashchenkodmitry@mail.ru

Present Address

Dmitry Pashchenko, Samara State
Technical University, Molodogvardeiskaya
str. 244, Samara, 443100, Russia

Summary

The process flow schematic of fuel-consuming equipment with thermochemical waste-heat recuperation by steam methane reforming with an addition of flue gas to the reaction mixture is suggested. The advantages of such a thermochemical recuperation (TCR) system compared with the TCR system by steam methane reforming are shown and justified. Based on the first law energy analysis, the heat inputs and outputs of the TCR system were determined. To determine the exhaust gases heat transformed into chemical energy of a new synthetic fuel, the thermodynamic analysis by minimizing Gibbs energy via Aspen HYSYS was performed. It was found that with an increase in the mole fraction of combustion products in the reaction mixture, the enthalpy of the methane reforming reaction increases, especially noticeable at the temperature range above 1000 K. Based on the heat balance of the TCR system was established that the addition of combustion products to the reaction mixture has the following effects: reducing the heat input for steam production in a steam generator; reduction of the steam generator size because of the need to produce a smaller amount of steam in comparison with TCR by pure steam methane reforming; and reducing the amount of heat transferred through the wall of the reformer and, as a consequence, reduction in size of the reformer.

KEYWORDS

energy efficiency, flue gas, heat balance, thermochemical recuperation, waste-heat

1 | INTRODUCTION

According to the International Energy Agency report, the rate of increase in global energy demand has increased by approximately 2.1% in 2017, which is more than double the rate of increase in 2016, because of the global population and economic growth.¹ More than 70% of the energy consumption in the world is met by different fossil fuels, while the remaining 30% is covered by renewable energy and nuclear energy.² Natural gas supplies 22% of the energy used worldwide, and makes up nearly a quarter of electricity generation as well as playing a crucial role as a

feedstock for industry. Natural gas is a versatile fuel and its growth is linked in part to its environmental benefits relative to other fossil fuels, particularly for air quality as well as greenhouse gas emissions. Global demand for natural gas is forecast to increase at an average 1.6% over the next 5 years. According to the report of the International Energy Agency, the 21st century can be considered the "Golden Age of Gas."¹

The continuous growth of fossil fuel consumption necessitates further enhancement of fossil fuel-consuming equipment. One of the main ways to improve the energy efficiency of fossil fuel (in particular natural gas) consum-

ing equipment (gas furnaces, internal combustion engines [ICE], gas turbines, etc) is the recovery of flue gas heat.³ For example, according to the report of the Russian Ministry of Energy, in Russia, the average energy efficiency of steel-making furnaces is about 31%, forging furnaces is about 46%, and glass-melting furnaces is about 45%.⁴ The main reason for the low energy efficiency of such equipment is the large heat loss of waste flue gases.

In recent years, scientist and engineers are paying more attention to novel ways of the heat loss recuperation schemes. In this case, the recuperation system by thermochemical principles could be mentioned.^{5,6} These systems of exhaust heat recuperation are called thermochemical recuperation (TCR). TCR can be a perspective method improving the energy efficiency of various fuel-consuming equipment. The main concept of TCR is a transformation of exhaust heat into chemical energy of a new synthetic fuel that has higher calorimetric properties such as low heating value. In other words, TCR of exhaust heat occurs when chemical endothermic reactions are performed, for example, methane reforming. Knoche et al⁷ presented that the combustion irreversibility of synthesis gas is less than is of natural gas.

In the last years, a set of papers was published on thermochemical recuperation in various fuel-consuming equipment: gas turbines,⁸ industrial furnaces,⁹ internal combustion engines,¹⁰ etc.¹¹ Moreover, various ways of thermochemical recuperation performing were considered, eg, by steam methane reforming (SMR),^{12,13} steam reforming of ethanol,⁵ and reforming of isooctane.¹⁴ Results about the advantages of thermochemical recuperation vary extensively because many various schematic diagrams have been investigated.

Historically, the thermochemical waste-heat recuperation process by steam methane reforming is finding a wide application for industrial furnaces and gas turbines. Popov et al.⁹ reported that TCR of exhaust heat by steam methane reforming can be a promising way to improving energy efficiency of high-temperature plants (HTP), which use natural gas, for example, glass melting furnaces. The authors presented that HTP with thermochemical recuperation can reach saving up to 10% to 25% of fuel consumption without affecting other parameters of the process. Also, Maruoka et al¹⁵ investigated the steel-making furnace with thermochemical waste-heat recuperation by steam methane reforming. The authors reported the results of an exergy analysis and an economic evaluation of the use of TCR. The results showed that the exergy loss in the proposed system was only 15% of the total exergy losses in the conventional system.

Verkhivker and Kravchenko analyzed the energy efficiency of the combined cycle (gas turbine with steam cycle) with the TCR system by steam methane reforming.⁸ The

authors concluded that the energy efficiency of the combined gas turbine units with TCR can reach 80% to 90%. Also, the authors reported that the exergy efficiency of the gas turbine unit (GTU) with TCR has reached the level of 85% to 90%. For TCR by steam methane reforming, the overall lower heating value (LHV) of the new synthetic fuel can be 1.25 times higher than LHV of the original methane.⁸

The energy analysis of the thermochemical recuperation system by steam methane reforming showed that about 15% to 20% of heat in the TCR system is spent for the steam generation and preheating to the reforming temperature.¹² Moreover, chemically treated water must be used for the steam generation. Provided that the SMR reaction is usually carried out at a steam-to-methane ratio of 2:1, the part of the chemically treated water, and as a consequence, some of the steam is lost to the environment thereby reducing the cost-effectiveness of using the TCR systems. For example, to reform 1 m³ of methane by SMR even at a CH₄:H₂O=1:1 ratio, it is necessary to expend about 3000 kJ of heat in order to heat water from a temperature of 20° C and to obtain steam with temperature of 300° C (at p = 5bar). One of the ways to reduce the heat input for the production and heating of steam is to add a portion of the high-temperature flue gases to the reaction mixture of steam and methane. It is known that products of complete combustion of natural gas with air are containing approximately: 18% of steam, 9% of carbon dioxide, 71% of nitrogen, and 2% of other gases. Therefore, the TCR system will be using not only the heat of flue gases but the component composition of it, in particular, steam and carbon dioxide.

In this paper, a thermochemical recuperation system by steam methane reforming with an addition of flue gases is proposed and analyzed. The effect of flue gas addition on the energy efficiency of the TCR systems is investigated. The heat balance of the TCR system is compiled to determine the heat input and output.

2 | SCHEMATIC CONFIGURATION

The main concept of thermochemical recuperation of exhaust gas heat in the industrial gas furnace is the increase in the LHV of the new synthetic fuel (synfuel) compared with the original fuel converting the exhaust heat into chemical energy of a new synfuel. In conventional fuel-consuming equipment, in particular, industrial gas furnaces, natural gas is combusted directly in the combustion chamber. For example, when 1 m³ of methane is completely combusted, 35.82 kJ is produced. In the furnaces with thermochemical recuperation, natural gas is primarily reformed to syngas in the chemical reactor (methane reformer). For example, when burning syngas

generated by full steam reforming of 1 m³ of methane, 45.03 kJ of heat is produced. In summary, we can say that in the fuel-consuming equipment with thermochemical recuperation of exhaust gas heat the heat generation from the original fuel in a combustion chamber is divided into two stages. The first stage is the increasing of the LHV of the original fuel by the transformation of exhaust gas heat into the chemical energy of new synthetic fuel. And the second stage is the combustion of new synfuel that has higher LHV than that of the original fuel. For example, (a) the schematic diagram of heat extraction from the original fuel for the conventional fuel-consuming equipment and (b) for the fuel-consuming equipment with thermochemical recuperation by steam methane reforming are depicted in Figure 1.

As mentioned above, the steam production for the TCR system is requiring a large amount of heat. For example, to produce 1 kg of saturated steam at 5 bar from liquid water with temperature of 70° C, 2454.8 kJ of heat must be supplied. And to heat the produced saturated steam from saturation temperature to 600° C, 954.35 kJ must be additionally supplied. Methane before the reforming process should be preheated. The heat exchangers are used for steam and methane heating before SMR.¹⁶

Usually, in order to produce steam, to heat the reaction mixture of steam and methane to temperatures of 500 to 600° C, and to completely reform of 1 m³ of methane, it is necessary to have the temperature of the exhaust flue gases more than 1000° C (in this example, the exhaust flue gases are obtained by combustion of 1 m³ of methane). If the temperature of flue gases is below 1000° C, then an additional heat source should be installed, for example, an autonomous steam generator. Therefore, in that case, thermochemical regeneration, although it will have a positive energy effect, will have a negative economic efficiency.

The problem of high heat input for steam production in the TCR systems can be solved by the addition of combustion products to the reaction steam/methane mixture. In this case, part of the heat of the high-temperature flue gases will be transferred to the reaction steam/methane mixture almost instantly in the mixing ejector. In addition, according to the chemical equation of methane combustion, about 29% of oxidizing agents (steam and carbon dioxide) are contained in flue gases. Therefore, in this case, the blend composition of flue gases, in particular, steam and carbon dioxide, will also be used in the TCR system. The process flow schematic of a gas furnace with ther-

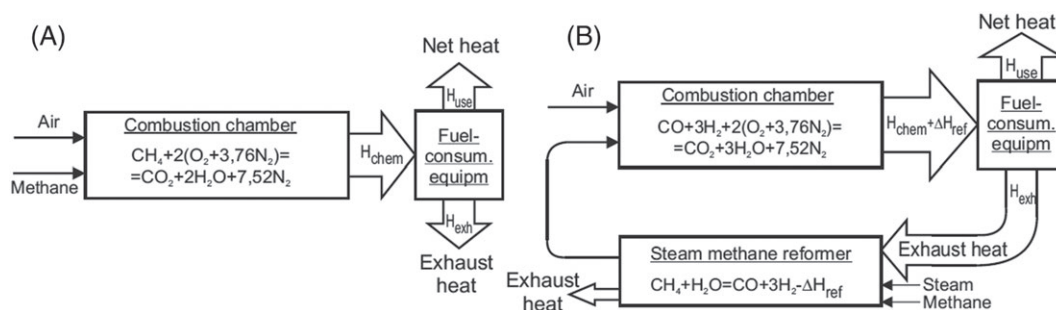


FIGURE 1 A, The schematic diagram of heat extraction from the original fuel for the conventional fuel-consuming equipment and B, for the fuel-consuming equipment with thermochemical recuperation by steam methane reforming

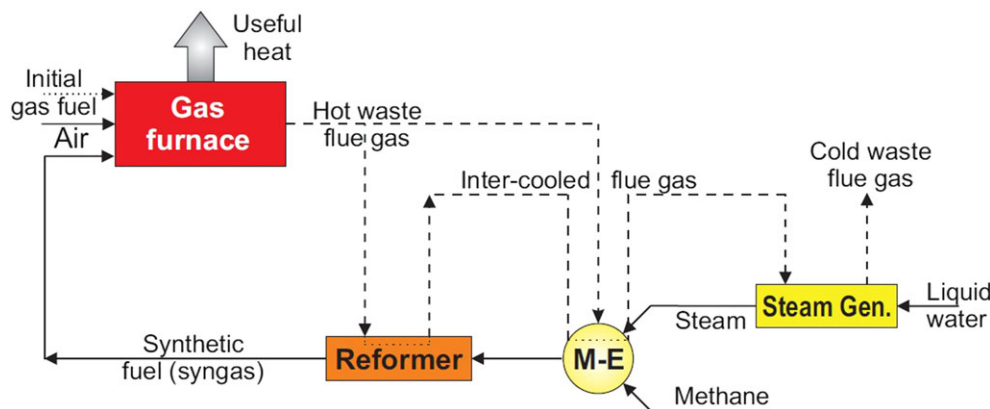
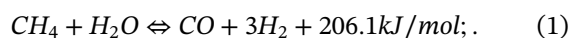


FIGURE 2 The process flow schematic of a gas furnace with thermochemical waste-heat recuperation by steam methane reforming with addition of flue gas to the reaction mixture [Colour figure can be viewed at wileyonlinelibrary.com]

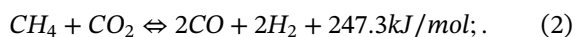
mochemical waste-heat recuperation by steam methane reforming with the addition of flue gas to the reaction mixture is presented in Figure 2.

The process flow schematic in Figure 2 can be virtually divided in two subsystems. The first is the heat generation and consumption subsystem with main components such as a combustion chamber and burner. The second is the thermochemical exhaust heat recuperation subsystem with main components such as a methane reformer, a mixer-heat exchanger (M-E), and a steam generator. Usually, the reaction mixture before a methane reformer is preheated to 400 to 600° C. To preheat the reaction mixture, the inter-cooled flue gas is used. The flue gases after a gas furnace are divided in two streams. The first flue gas stream is sequentially passing through a methane reformer, a mixer-heat exchanger and a steam generator. The second flue gas stream is passing into the mixer-heat exchanger where steam and methane are also fed. After mixing, the reaction mixture (methane, steam, and combustion products) is fed into the reaction space of the methane reformer. The reformer is a shell-and-tube heat exchanger where the exhaust gas heat of the first flue gas stream is transferred through the wall to the reaction space.^{17,18} One side of the wall is filled with the first stream of exhaust gases, the other side is filled with the reaction mixture. In the methane reformer, methane, steam, and carbon dioxide (from combustion products) are reformed due to endothermic reaction: (a) steam methane reforming and (b) dry methane reforming (DMR), to the synthesis gas that is called the synthetic fuel or synfuel. The reaction space of the reformer is activated by the nickel-based¹⁹ or other catalysts.^{20,21}

Steam methane reforming as



DMR as



The methane reforming processes are absorbing a large amount of the high-temperature thermal energy that is mainly extracted through the wall from the first stream of flue gases after the gas furnace. In addition, part of the heat for the methane reforming process is supplied with the reaction mixture. The synthetic fuel at the reformer outlet consists of the different gases (H_2 , CO , unreacted CH_4 , CO_2 , N_2 , and others). The syngas after the reformer is fed into the combustion chamber of the gas furnace. Steam for methane reforming is produced in the steam generator through the use of the residual exhaust heat.

The main advantages of the thermochemical recuperation scheme that presented in Figure 2 are the following:

- reducing the heat input for steam production in a steam generator;
- reduction of the steam generator size because of the need to produce a smaller amount of steam in comparison with TCR by pure steam methane reforming;
- reducing the amount of heat transferred through the wall of the reformer and, as a consequence, reducing the specific amount of metal for the reformer; and
- combustion hydrogen-rich fuel instead of natural gas.

In addition, as an example of the advantages of the scheme in Figure 2, it is necessary to emphasize the fact that the rate of heating of the reaction mixture before the reaction is increasing. This is because of the fact that the combustion products transfer heat to the reaction mixture by mixing. Rapid heating of the mixture is increasing the efficiency of the reformer. Since when the reaction mixture is slowly heated, there is an intensive methane cracking reaction, in the temperature range above 300° C. Because of slow heating, soot carbon may be contained in the reaction mixture, which reduces the catalytic activity in the reformer.

3 | METHODOLOGY

The main aim of this investigation is the determination of the heat inputs and outputs of the thermochemical recuperation system that is presented in Figure 2. Moreover, the effect of flue gas addition on the energy efficiency of the TCR systems is investigated. The schematic view of the heat inputs and outputs of the TCR system is depicted in Figure 3.

This schematic view in Figure 3 has the following notations: H_{air} —enthalpy of air for combustion; $H_{\text{fg}(I)}$ and $H_{\text{fg}(II)}$ —enthalpy of first and second stream of flue gases after a furnace, respectively; H_{fg}' —enthalpy of inter-cooled flue gases; H_{loss} —waste-heat loss; H_{st} —enthalpy of steam after a steam generator; H_{mix} —enthalpy of the reaction mixture after a mixer-heat exchanger (M-E); H_{CH_4} —enthalpy of initial methane; H_{water} —enthalpy of feed water at 70° C; H_{sf} —enthalpy of synfuel after a reformer; and $-H_{\text{ref}}$ —enthalpy of the methane reforming reaction.

The problem of determining the heat outputs in the TCR system is a conjugate problem with the thermodynamic analysis of the methane reforming process. This is because of the fact that the synthetic fuel composition, the methane conversion, the enthalpy of the methane reforming reaction, etc, depending on the operational conditions such as temperature, pressure, and composition of the inlet reaction mixture. The interconnection of these parameters allows the establishing of a thermodynamic analysis.^{22,23}

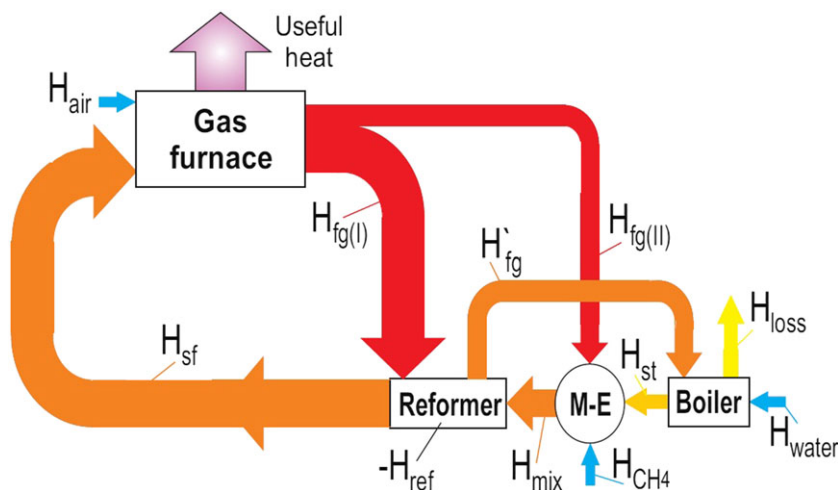


FIGURE 3 The schematic view of the heat inputs and outputs of the thermochemical recuperation (TCR) system [Colour figure can be viewed at wileyonlinelibrary.com]

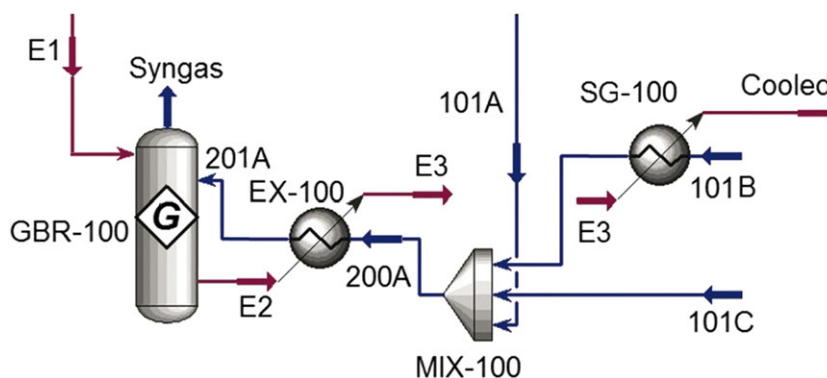


FIGURE 4 Aspen HYSYS process flow diagram for thermochemical recuperation (TCR) by steam methane reforming with addition of combustion products to the reaction mixture [Colour figure can be viewed at wileyonlinelibrary.com]

The main goal of the thermodynamic analysis is the calculation of synthetic fuel compositions at various operating conditions such as temperature, pressure, and inlet gas composition. The thermodynamic analysis makes it possible to determine the limiting (equilibrium) synfuel compositions by using the method of Gibbs free energy minimization (GFEM). The basic concept of GFEM is built on the principle that the thermochemical system is thermodynamically favorable when its total Gibbs energy is at its minimum value and its differential is zero for each pressure and temperature.^{24–26} The basic concept of GFEM can be written as

$$(dG^t)_{T,p} = 0. \quad (3)$$

The algorithm of the determination of total Gibbs energy was discussed in the author's previous work.²⁴ The resulting equation for determination of total Gibbs energy is defined as follows:

$$G^t = \sum n_i \Delta G_{fi}^0 = \sum n_i RT \ln \left(\frac{f_i}{f_i^0} \right), \quad (4)$$

where G^t is the total Gibbs free energy as a function on a temperature; n_i is the number of moles of element i ; G_{fi}^0 is the standard Gibbs energy of formation of element i ; R is the molar gas constant; T is the temperature; f_i is the fugacity of pure element i ; and f_i^0 is fugacity of pure element i at the standard state.

In this paper, the thermodynamic analysis by using the GFEM technique is performed by using Aspen HYSYS. The Aspen HYSYS software contains an extensive database of thermodynamic properties of chemical components, algorithms for the database computing, and program, which allow the determination of equilibrium composition and thermodynamic properties of the chemical system to be examined.^{27,28} Aspen HYSYS is calculating the distribution of reaction products by minimizing the Gibbs energy of each species in the reaction system regardless of the reaction network. This program is used for the calculation of fundamental principles of the law of mass action and the law of mass and energy conservation.

The following assumptions have been made in the thermodynamic analysis:

1. Steady state operation.
2. Isothermal conditions prevail.
3. Negligible pressure drop.
4. Plug flow in the virtual reactor.
5. No interphase and intraparticle mass transfer limitations.
6. Flue gas composition is fixed.
7. No heat loss through the reformer wall.

The calculation Aspen HYSYS model is presented in Figure 4.

In the TCR systems where the oxidant-to-methane ratio is higher than 1, the syngas contains a residual oxidizer. In this case, as the oxidant-to-methane ratio increases, the methane conversion will increase but the overall energy efficiency of the TCR system will decrease because synfuel will contain a large proportion of noncombustible components. The synfuel will also contain nitrogen when the addition of combustion products to the reaction mixture will take place.

4 | RESULTS AND DISCUSSION

4.1 | Chemical reactions

The overall reaction in the methane reformer to form H_2 , CO , CO_2 , H_2O , and other components may include the equations listed in Table 1.²⁹ In addition, other chemical reactions may occur in the reformer, which is basically a combination of the reactions shown in the Table 1. However, it is well known that for the thermodynamic analysis, it is sufficient to consider only elementary reactions presented in Table 1.

Figure 5 shows the dependence of the change of Gibbs free energy versus temperature for reactions presented in Table 1.

TABLE 1 Possible chemical reactions in the reformer^{29,33}

No.	Reaction	ΔH_{298} , (kJ/mol)
Methane oxidation		
1	$CH_4 + H_2O \rightleftharpoons CO + 3H_2$	206.4
2	$CO + H_2O \rightleftharpoons CO_2 + H_2$	-41.1
3	$CH_4 + CO_2 \rightleftharpoons 2H_2 + 2CO$	247.3
Carbon formation		
4	$CH_4 \rightleftharpoons C + 2H_2$	74.8
5	$2CO \rightleftharpoons C + CO_2$	-173.3
6	$CO + H_2 \rightleftharpoons C + H_2O$	-90.13
7	$CO_2 + 2H_2 \rightleftharpoons C + 2H_2O$	-131.3

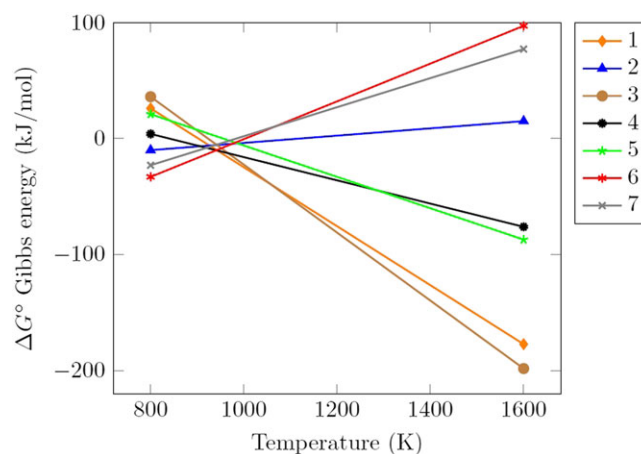


FIGURE 5 The Gibbs free energy as a function of temperature for reactions in Table 1 [Colour figure can be viewed at wileyonlinelibrary.com]

As it can be seen from Figure 5, the Gibbs free energy for oxidation reactions (1 and 3 from Table 1) decreases significantly with increasing temperature. In the temperature range above 900 K, reforming reactions (1 and 3 from Table 1) are more likely to be thermodynamic than the reactions of methane cracking (4) and Boudouard reaction (5).³⁰ At temperatures above 1000 K, the equilibrium of the reactions (6 and 7) is shifted toward the formation of gasification products of the carbon. Therefore, it can be concluded that the reactions of steam and carbon dioxide reforming mainly take place in the methane reformer.

Carbon deposition is another factor that needs to be considered especially at high temperatures. However, there is sufficient information in literature that carbon formation can be controlled by improving the coking resistance of the catalysts as well as by regulating the operating conditions such as steam-to-carbon and fuel-to-oxidant ratio.³¹ In this study, oxidizer-to-carbon ratio in all different feed gas compositions is either equal to or greater than unity, which indicates that there is enough oxygen to oxidize and prevent formation of solid carbon.³² Hence, carbon formation was omitted from equilibrium expressions.

4.2 | Reaction enthalpy

According to the first law of thermodynamics, the enthalpy of flue gases transformed into the chemical energy of a synthetic fuel is determined by the enthalpy of the methane reforming reaction. The overall contribution to the material and energy balance of water-gas shift reaction is insignificant and may not be taken into account in thermodynamic analysis.²⁹ This disregard is because of the fact that in the temperature range of more than 600 K, the change in the free Gibbs energy for reactions (Equation 1

and 2) is substantially greater than the change in the Gibbs energy for water-gas shift reaction (pos. 2 in Table 1).

According to thermodynamic analysis, as the oxidizer-to-methane ratio in the reaction mixture increases, the methane conversion increases. However, an increase in the more stoichiometric oxidizer-to-methane ratio results in an increase in the volume fraction of the incombustible components in the synthesis gas. At the same time, at the stoichiometric ratio, the formation and deposition of the soot carbon are possible. The soot carbon leads to a decrease in the catalytic activity of the reformer. Based on the author's previous experimental study of methane conversion by steam and combustion products, it has been established that at a ratio of about 2 the activity of the catalyst (for example, nickel-based catalyst) is constant. Therefore, for further analysis, the oxidizer-to-methane ratio is 2. Moreover, for further analysis, the combustion products obtained by combustion of pure methane with an excess air ratio of 1 are considered.

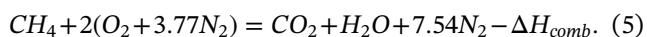


Figure 6 shows the enthalpy of the methane reforming reaction in the temperature range 600 to 1200 K at a pressure of 1 bar for various compositions of the initial reaction mixture. The axis of abscissa is showing the mol number of oxidizers that is containing in the combustion products. 0.0 is indicating that there are no combustion products in the reaction mixture. 1.0 is indicating that in the reaction mixture 1 mole of oxidizer from the combustion products, for example, $\frac{1}{3}(\text{CO}_2 + 2\text{H}_2\text{O} + 7.52\text{N}_2) = 1$ mol of oxidizer.

As can be seen from Figure 6, with increasing temperature, the reaction enthalpy increases. This is because of the fact that the CH_4 conversion is increasing. In addition,

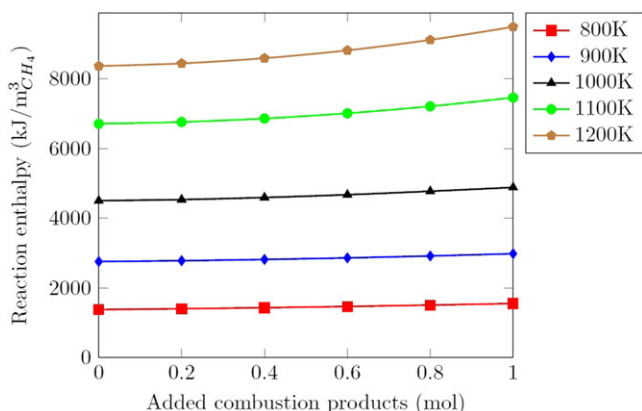


FIGURE 6 The enthalpy of the steam methane reforming (SMR) reaction as a function of temperature for various compositions of the initial reaction mixture β at $p=10$ bar [Colour figure can be viewed at wileyonlinelibrary.com]

with an increase in the proportion of oxidizers from the combustion products, the reaction enthalpy also increases. However, at the temperature range below 1000 K, the reaction enthalpy increases slightly. On the contrary, in the temperature range above 1000 K, the reaction enthalpy increases significantly. This is because of the fact that, in the temperature range below 1000 K, the SMR reaction is most likely to occur, and almost all of the methane reacts with steam, and carbon dioxide mostly remains unreacted. At the same time, at the temperature range above 1000 K, the change in Gibbs free energy for the DMR reaction is less than for the SMR reaction; therefore, carbon dioxide is consumed first in the reformer. In this case, the enthalpy of the DMR reaction is about 20% higher than the enthalpy of the SMR reaction.

Obviously, the energy efficiency of using the TCR system depends on the enthalpy of the methane reforming reaction. If the methane reforming reaction does not take place in the reformer, then the energy efficiency of the use of the TCR system is zero or even has a negative value, since, for steam production, exhaust heat is used. On the other hand, when the oxidizer-to-methane ratio is above stoichiometric, the synfuel is containing the noncombustible components, in particular, steam. In addition, with an increase in the proportion of combustion products in the reaction mixture, nitrogen is added as a noncombustible component of the synthesis gas. In this case, the noncombustion components although a part of the synthetic fuel do not positively participate in the combustion process from the energy point of view. Therefore, the noncombustion components are the ballast. On the other hand, the addition of combustion products to the reaction mixture reduces heat consumption for steam production. Therefore, the heat inputs and outputs are determined in the next section of this paper.

4.3 | Heat balance

To compile the heat balance of the TCR system that is shown in Figure 2 as well as to perform a first law energy analysis of various options (adding different amounts of flue gases to the reaction mixture) the following initial data were taken:

- The composition of flue gas (mole fraction of each element) after a furnace: $\text{CO}_2=0.095$; $\text{H}_2\text{O}=0.19$; and $\text{N}_2=0.715$; The flue gas mass after the furnace is 8.271 kg (obtained by complete combustion of 1 m³ of methane with air excess factor of 1.0).
- The temperature of feed water is 70° C, the temperature of methane before the mixer-heat exchanger is 20° C.
- The temperature of steam after the steam generator is 200° C ($p=3\text{bar}$).

- The reaction mixture is preheated in the M-E to 400° C (for some cases, the reaction mixture temperature is more than 400° C. In this case, the additional heat is taken into account in the heat balance of the reformer).
- The reforming temperature is 300° C less than the temperature of flue gas after the furnace;
- The synfuel temperature is equal to the reforming temperature.
- The oxidizer-to-methane ratio is 2 for all cases; 1 m³ of methane is added for the methane reforming process.

The following notation is used for the heat inputs and outputs: $H_{fg(I)}$ is the enthalpy of the first stream of exhaust flue gases after the furnace ($H_{fg(I)}$ in Figure 3), ΔH_{steam} is change in enthalpy of feed water in the steam generator, ΔH_{me} is the change in enthalpy of the reaction mixture in the mixer-heat exchanger, ΔH_{sf} is the change in enthalpy of the reaction mixture in the reformer, H_{ref} is enthalpy of the methane reforming reaction, and H_{loss} is enthalpy of the flue gases after the steam generator (For some cases, the H_{loss} is less than zero). This suggests that the enthalpy of the first flue gas stream is not enough to perform thermochemical recuperation in the given parameters that is mentioned above. To perform the TCR process in the given parameters, this requires the supply of heat from other sources. For example, the steam production and heating the reaction mixture is performed by using other sources of heat).

The energy analysis of the thermochemical recuperation system by steam methane reforming showed that about 15% to 20% of heat in the TCR system is spent for the steam generation and preheating to the reforming temperature.¹² Moreover, chemically treated water must be used for the steam generation. Provided that the SMR reaction is usually carried out at a steam-to-methane ratio of 2:1, the part of the chemically treated water, and as a consequence, some of the steam is lost to the environment thereby reducing the cost-effectiveness of using the TCR systems. For example, to reform of 1 m³ of methane by SMR even at a CH₄:H₂O=1:1 ratio, it is necessary to expend about 3000 kJ of heat in order to heat water from temperature of 20° C and to obtain steam with temperature of 300° C (at p=5bar). Therefore, the use of combustion products significantly reduces energy outputs for steam production.

Figure 7 shows the heat balance of the TCR system when the temperature of exhaust gas is 800° C.

In Figure 7, the axis of abscissa is the number of moles of the oxidizers that is containing in the second stream of the flue gases (according to Figure 3). Here, 0.2 mol of oxidizer is contained in 557.6 g of flue gas. As can be seen from Figure 7, with an increase in flue gas extraction into the second stream (for the reforming reaction), the enthalpy of

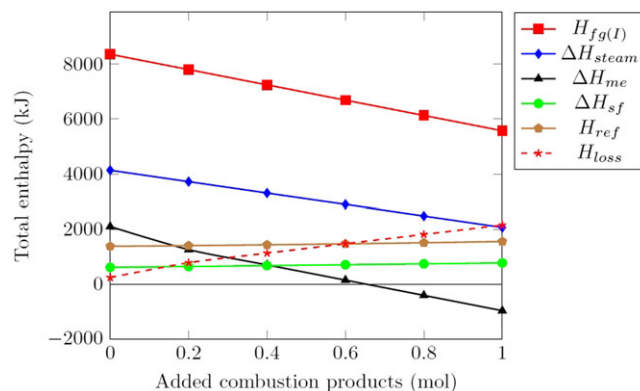


FIGURE 7 The heat balance of the thermochemical recuperation (TCR) system when the temperature of exhaust gas is 800° C [Colour figure can be viewed at wileyonlinelibrary.com]

the first flue gas stream decreases. The temperature of the reaction mixture after the mixer-heat exchanger increases with increasing flue gas extraction for the reaction (second stream in Figure 3). When the flue gas extraction is about 0.6 mol of oxidizers (1672.8 g of flue gases), the temperature of the reaction mixture is about 400° C and with increasing the flue gas extraction (more than 0.6 mol), the temperature of the reaction mixture increases. The heat for steam generation is decreasing when the flue gas extraction increases because the amount of pure steam after the steam generator decreases. On the other hand, the enthalpy of flue gases after the steam generator is positive for all variants. This means that there is enough heat to complete the TCR process. Additional heat supply from other sources is not required. However, it is worth noting that the enthalpy of the reforming reaction for a temperature of 773 K is minimal. Therefore, the main recuperation is achieved by heating the synfuel, and not by the transformation of the flue gas enthalpy into chemical energy of the new synthesis fuel. The reaction enthalpy has a significant value at temperatures above 1000 K.

Figure 8 depicts the heat balance of the TCR system when the temperature of exhaust gas is 1000° C. $H_{fg(I)}$ is monotone increasing with increasing temperature. ΔH_{steam} is constant for each flue gas extraction; with increasing flue gas extraction for the reforming reaction, ΔH_{steam} decreases due to a decrease in consumption of pure steam for the methane reforming process. However, other inputs and outputs of the heat balance are significantly different.

As can be seen from Figure 8, the reaction mixture reaches 400° C with 0.5 mol of added flue gas. An additional increase in the amount of flue gas for the methane reforming reaction increases the reaction mixture temperature. The additional enthalpy of the reaction mixture after the mixer-heat exchanger reduces the change of enthalpy of the synthetic fuel in the reformer. This is clearly seen

in Figure 8, the enthalpy decreases with the increase of the flue gas extraction above 0.5 mol. Moreover, Figure 8 makes it clear that the enthalpy of the first flue gas stream is not enough to perform TCR in the given parameters.

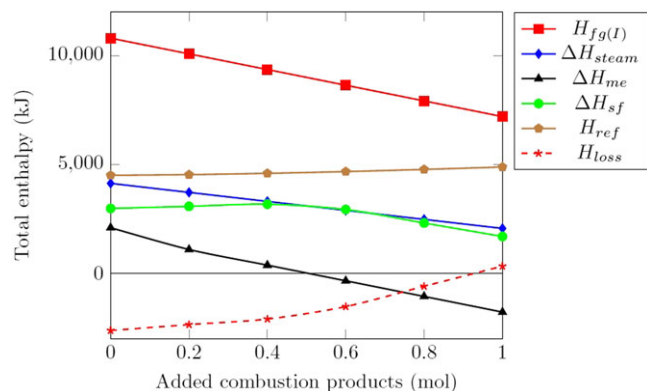


FIGURE 8 The heat balance of the thermochemical recuperation (TCR) system when the temperature of exhaust gas is 1000° C [Colour figure can be viewed at wileyonlinelibrary.com]

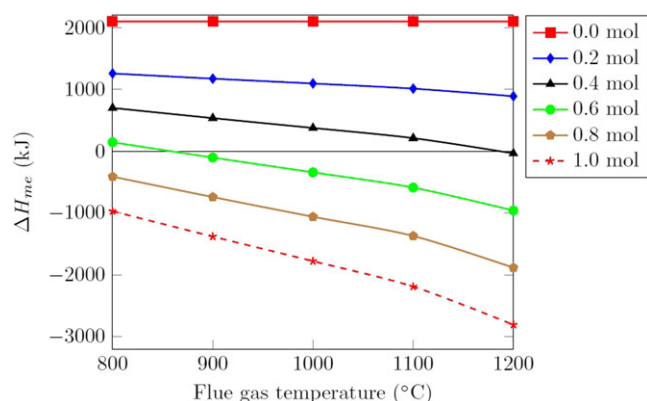


FIGURE 9 The change of the mixture enthalpy in the mixer-heat exchanger (ΔH_{me}) vs flue gas temperature [Colour figure can be viewed at wileyonlinelibrary.com]

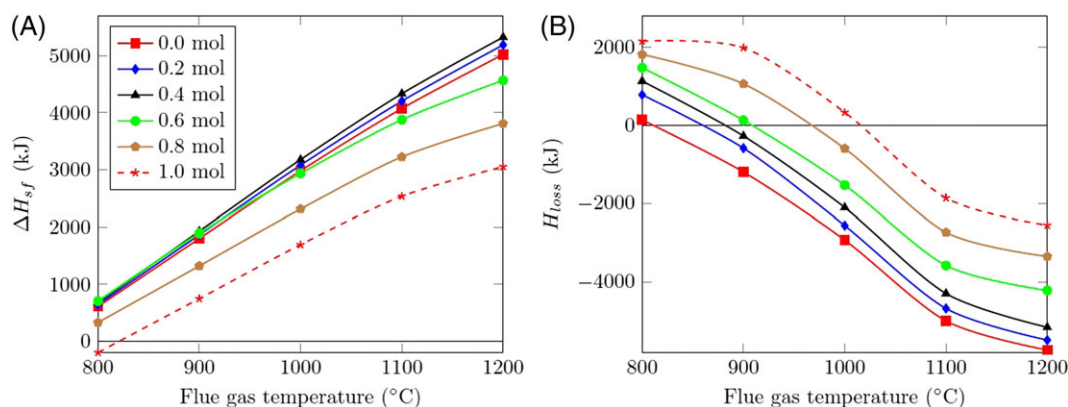


FIGURE 10 A, The change of the synfuel enthalpy in the reformer (ΔH_{sf}) and B, the heat loss (H_{loss}) vs flue gas temperature [Colour figure can be viewed at wileyonlinelibrary.com]

A heat deficit occurs for the entire range under study. However, with an increase in the flue gas extraction, the heat deficit decreases, because of the consumption of pure steam decreases and the heat consumption for heating the synthesis fuel in the reformer decrease.

The main advantage of the TCR system with the addition of flue gases to the reaction mixture is the decreasing of heat consumption for production of the fuel oxidizers and heating of reaction mixture. This advantage is clearly presented in Figure 9. Figure 9 shows that with increasing flue gas extraction for the methane reforming reaction, the amount of heat required to preheat the reaction mixture decreases. In addition, it is worth considering the fact that with an increase in flue gas extraction, the amount of heat consumed to produce pure steam also decreases. Adding flue gases makes it possible to remove the mixer-heat exchanger in the scheme in Figure 9.

Figure 10 depicts (a) the change of the synfuel enthalpy in the reformer and (b) the heat loss versus flue gas temperature. As can be seen from Figure 10A that the addition of combustion products significantly reduces the amount of heat required for heating the synfuel in the reformer. Reducing the consumption of heat for heating the synfuel in the reformer by 35% to 55% makes it possible to reduce the heat exchange surface of the reformer.

Figure 10B shows that in the temperature range above 1000° C, the enthalpy of the first flue gas stream after the furnace is not enough to perform TCR. Therefore, there is a heat deficit in the TCR system. However, the addition of flue gases to the reaction mixture reduces this heat deficit. It is expected that a further increase in the amount of combustion products will remove the heat deficit in the TCR system.

5 | CONCLUSION

The results of this paper are confirming the main advantages of the TCR system by steam methane reforming with flue gas addition:

- reducing the heat input for steam production in a steam generator;
- reduction of the steam generator size due to the need to produce a smaller amount of steam in comparison with TCR by pure steam methane reforming;
- reducing the amount of heat transferred through the wall of the reformer, and, as a consequence, reducing the specific amount of metal for the reformer; and
- combustion hydrogen-rich fuel instead of natural gas.

Adding approximately 1.38 kg of combustion products with a temperature of 1000° C to the reaction mixture in such a way as to the oxidizer-to-methane ratio was 2:1, it is possible to refuse the reaction mixture preheater in the TCR system, and the consumption of pure steam is reduced by 25%. When approximately 2.750 kg of combustion products with a temperature of 1000° C are added to the reaction mixture, the TCR system does not have a heat deficit and the recovery rate of the TCR system is about unity (in terms of the equivalent amount of 1 m³ of inlet methane).

ORCID

Dmitry Pashchenko  <https://orcid.org/0000-0003-0249-6137>

REFERENCES

1. Birol F, Besson C. Golden rules for a golden age of gas - 2012, world energy outlook special report on unconventional gas, Paris, France, International Energy Agency; 2012.
2. Office GP. *International Energy Outlook 2016: With Projections to 2040*. Washington, DC: Government Printing Office; 2016.
3. Fiaschi D, Manfrida G, Russo L, Talluri L. Improvement of waste heat recuperation on an industrial textile dryer: Redesign of heat exchangers network and components. *Energy Convers Manage*. 2017;150:924-940.
4. Makarov A, Chupyatov V. Energy efficiency and energy saving in Russia. *Int J Global Energy*. 2016;39(3):2000:1-3.
5. Tartakovsky L, Sheintuch M. Fuel reforming in internal combustion engines. *Prog Energy Combust Sci*. 2018;67:88-114.
6. Pashchenko D. Thermochemical recovery of heat contained in flue gases by means of bioethanol conversion. *Therm Eng*. 2013;60(6):438-443.
7. Harvey S, Knoche K, Richter H. Reduction of combustion irreversibility in a gas turbine power plant through off-gas recycling. *J Eng Gas Turbines Power*. 1995;117(1):24-30.
8. Verkhivker G, Kravchenko V. The use of chemical recuperation of heat in a power plant. *Energy*. 2004;29(3):379-388.
9. Popov S, Svistunov I, Garyaev A, Serikov E, Temyrkanova E. The use of thermochemical recuperation in an industrial plant. *Energy*. 2017;127:44-51.
10. Poran A, Thawko A, Eyal A, Tartakovsky L. Direct injection internal combustion engine with high-pressure thermochemical recuperation-Experimental study of the first prototype. *Int J Hydrogen Energy*. 2018;43(27):11969-11980.
11. Liu D, Xin-Feng L, Bo L, Si-quan Z, Yan X. Progress in thermochemical energy storage for concentrated solar power: a review. *Int J Energy Res*. 2018;42(15):4546-4561.
12. Pashchenko D. First law energy analysis of thermochemical waste-heat recuperation by steam methane reforming. *Energy*. 2018;143:478-487.
13. Rostrup-Nielsen J. Steam reforming and chemical recuperation. *Catal Today*. 2009;145(1-2):72-75.
14. Chakravarthy VK, Daw CS, Pihl JA, Conklin JC. Study of the theoretical potential of thermochemical exhaust heat recuperation for internal combustion engines. *Energy Fuels*. 2010;24(3):1529-1537.
15. Maruoka N, Mizuochi T, Purwanto H, Akiyama T. Feasibility study for recovering waste heat in the steelmaking industry using a chemical recuperator. *ISIJ Int*. 2004;44(2):257-262.
16. Yuan Q, Gu R, Ding J, Lu J. Heat transfer and energy storage performance of steam methane reforming in a tubular reactor. *Appl Therm Eng*. 2017;125:633-643.
17. Ngo SI, Lim Y-I, Kim W, Seo DJ, Yoon WL. Computational fluid dynamics and experimental validation of a compact steam methane reformer for hydrogen production from natural gas. *Appl Energy*. 2019;236:340-353.
18. Kumar A, Baldea M, Edgar TF. A physics-based model for industrial steam-methane reformer optimization with non-uniform temperature field. *Comput Chem Eng*. 2017;105:224-236.
19. Pashchenko D. Effect of the geometric dimensionality of computational domain on the results of CFD-modeling of steam methane reforming. *Int J Hydrogen Energy*. 2018;43(18):8662-8673.
20. Ding H, Xu Y, Luo C, et al. A novel composite perovskite-based material for chemical-looping steam methane reforming to hydrogen and syngas. *Energy Convers Manage*. 2018;171:12-19.
21. Musamali R, Isa YM. A novel catalyst system for methane decomposition. *Int J Energy Res*. 2018;42(14):4372-4382.
22. Özkara-Aydinoğlu Ş. Thermodynamic equilibrium analysis of combined carbon dioxide reforming with steam reforming of methane to synthesis gas. *Int J Hydrogen Energy*. 2010;35(23):12821-12828.
23. Tabrizi FF, Mousavi SAHS, Atashi H. Thermodynamic analysis of steam reforming of methane with statistical approaches. *Energy Convers Manage*. 2015;103:1065-1077.
24. Pashchenko D. Thermodynamic equilibrium analysis of combined dry and steam reforming of propane for thermochemical waste-heat recuperation. *Int J Hydrogen Energy*. 2017;42(22):14926-14935.
25. Baamran KS, Tahir M. Thermodynamic investigation and experimental analysis on phenol steam reforming towards enhanced H₂ production over structured ni/zntio₃ nanocatalyst. *Energy Convers Manage*. 2019;180:796-810.
26. Aktaş S, Karakaya M, Avcı AK. Thermodynamic analysis of steam assisted conversions of bio-oil components to synthesis gas. *Int J Hydrogen Energy*. 2009;34(4):1752-1759.
27. Øi LE. Aspen HYSYS simulation of CO₂ removal by amine absorption from a gas based power plant. In: The 48th Scandinavian Conference On Simulation And Modeling (SIMS 2007). Linköping University Electronic Press; 2007; Göteborg:73-81.

28. Feyzi V, Beheshti M, Kharaji AG. Exergy analysis: A CO₂ removal plant using a-MDEA as the solvent. *Energy*. 2017;118:77-84.
29. Xu J, Froment GF. Methane steam reforming, methanation and water-gas shift: I. intrinsic kinetics. *AIChE J*. 1989;35(1):88-96.
30. Wang Z, Fan W, Zhang G, Dong S. Exergy analysis of methane cracking thermally coupled with chemical looping combustion for hydrogen production. *Appl Energy*. 2016;168:1-12.
31. Choudhary VR, Mondal KC. CO₂ reforming of methane combined with steam reforming or partial oxidation of methane to syngas over NdCoO₃ perovskite-type mixed metal-oxide catalyst. *Appl Energy*. 2006;83(9):1024-1032.
32. Yang M, Papp H. CO₂ reforming of methane to syngas over highly active and stable Pt/MgO catalysts. *Catal Today*. 2006;115(1-4):199-204.
33. Hoang D, Chan S, Ding O. Kinetic and modelling study of methane steam reforming over sulfide nickel catalyst on a gamma alumina support. *Chem Eng J*. 2005;112(1):1-11.

How to cite this article: Pashchenko D. Thermochemical waste-heat recuperation by steam methane reforming with flue gas addition. *Int J Energy Res*. 2019;1-11. <https://doi.org/10.1002/er.4436>